

Amogh Parab

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SUMMARY

Mathematics PhD from The Ohio State University with more than three years of experience in both object-oriented programming (Python) and functional programming (Lean4) • **Contributed to open-source projects** and currently formalizing mathematical research problems to **support AI training pipelines at Harmonic** • Passionate about formal methods and their applications to scientific discovery and AI safety.

SKILLS

- Formal Methods: Lean4 (Mathlib, Verified-zkEVM-CompPoly), dependent type theory, formal verification, theorem proving
- Languages: Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), LaTeX, Julia
- Mathematics: Category Theory, Algebraic Topology, Abstract Algebra, Type Theory, Numerical Methods
- OS and Platforms: Bash, GitHub, Windows, Linux, Jupyter Notebook, Visual Studio Code

WORK EXPERIENCE

Harmonic

Remote

Lean Expert (part-time)

Sep. 2025 – Present

- **Formalized over 200 advanced mathematics problems** in Lean4, spanning a broad range of topics, including algebra, geometry, and number theory to support the development of an advanced mathematical reasoning model.
- **Contributed to open-source formalization projects**, including Mathlib, Google DeepMind's Formal Conjectures, and Verified-zkEVM-CompPoly, expanding the frontier of machine-verifiable mathematics.
- Designed and implemented a Euclidean Geometry library in Lean4 to streamline formalization workflows.

The Ohio State University

Columbus, Ohio

Graduate Teaching Associate / Lecturer

Aug. 2020 – May 2026

- Created and presented over **150 hours of mathematics content**, incorporating visual demonstrations to enhance student engagement, with an average teaching score of 4.7/5.
- Led Engineering Mathematics recitations, conducted office hours, proctored exams, and prepared teaching materials for undergraduate engineering students.

PROJECT EXPERIENCE

Rewriting Systems in Lean4 ([GitHub](#))

Columbus, Ohio

The Ohio State University

Aug. 2023 – Jan. 2025

- **Designed an inductive data type in Lean4** to serve as a rewriting system for multiplicative words, a core component of the categorical group structures in my PhD thesis. Extended Mac Lane's coherence theorem to categorical groups.
- Formally verified fundamental algebraic properties, including **associative concatenation and the universal property**; authored over **2,000 lines of Lean4 code** and **20+ pages of technical documentation in LaTeX**.

Formalizing Homotopy Lifting Property ([GitHub](#))

Berkeley, California

SLMath (formerly MSRI)

Jun. 2023

- **Collaborated with Mathlib maintainers** at SLMath to port the Lean mathematics library from Lean3 to Lean4 during a two-week formalization summer school.
- **Formalized the uniqueness of the homotopy lifting property** in Lean4, writing over 300 lines of code; presented findings alongside a team of six at the closing session.

Data Science Bootcamp (Project 1: [GitHub](#)) (Project 2: [GitHub](#))

Remote

Participant at the Erdős Institute

Sep. 2024 – Nov. 2024

- Predicted results of Valorant online matches with **67% accuracy using XGBClassifier** on selected key features from 160 engineered features; applied **grid search for hyperparameter tuning**.
- Developed a predictive model for car prices with an R^2 value of 0.88 using XGBoost in a team of four, **achieving a 22% improvement** over the baseline model; utilized SHAP values to identify key feature importance.

Creep Evolution in FCC Crystalline Structure

Texas A&M University (Internship)

College Station, Texas

May 2017 – Jul. 2017

- Analyzed stress-strain relationship of exposed creep on external boundary conditions using ABAQUS CAD software.

Quasi-Solution Approach to Non-Linear Differential Equations

The Ohio State University (Internship)

Columbus, Ohio

Aug. 2016 – Jan. 2017

- Provided analytic solution to the Blasius boundary condition (Fluid Dynamics) using Maple software. Used the Banach fixed-point theorem to show the existence of the solution.

Optimal Stochastic Growth Model Using U.S. Data (Penn World Table) ([GitHub](#))

Personal Project

Columbus, Ohio

Nov. 2025 – Dec. 2025

- Implemented a **discrete-time stochastic optimal growth model in Julia** using real macroeconomic data from the Penn World Table (PWT). Compared the theoretical optimal policy to observed outcomes.

PUBLICATIONS

- “Coherence and Symmetrization of Categorical Groups”, PhD thesis, The Ohio State University, 2025
- “Gröbner bases of Rational Normal Curves”, Proceedings of the COCOA Workshop, IIT Gandhinagar, 2019

GRANTS AND AWARDS

- Special Graduate Assignment for Thesis Research, The Ohio State University, 2023
- Bronze Medal for the best performance in the core courses of Mathematics, IIT Gandhinagar, 2019
- Dean’s List, IIT Gandhinagar, Spring 2015, Fall 2015, Spring 2016

EDUCATION

The Ohio State University

Columbus, Ohio

PhD in Mathematics (Cumulative GPA: 3.99/4)

Aug. 2025

Thesis: Coherence and Symmetrization of Categorical Groups

Coursework: Mathematics of Data Science, Abstract Algebra, Analysis, Lie Theory, Topological Quantum Field Theory

Indian Institute of Technology Gandhinagar

Gandhinagar, India

BTech in Mechanical Engineering, MSc in Mathematics (Cumulative GPA: 3.90/4)

Jul. 2019

Master’s Thesis: On Riemann-Roch Theorem

Coursework (Master’s): Linear Algebra, Numerical Methods, Ordinary and Partial Differential Equations, Group Theory

Coursework (BTech): Quantum Mechanics, Non-Linear Dynamics

LEADERSHIP EXPERIENCE

Directed Reading Program

Columbus, Ohio

Program Coordinator, The Ohio State University

Aug. 2021 – May 2025

- Organized the pairing process of more than **100 undergraduate mathematics enthusiasts** with experienced graduate mentors to explore topics of mutual interest.
- Mentored five students over two years** in advanced mathematical concepts beyond the standard curriculum; introduced formal theorem proving via Lean’s **Natural Number Game**.

Team LaTeX

Gandhinagar, India

Co-founder, IIT Gandhinagar

Aug. 2017 – May 2018

- Co-founded a LaTeX club** at IIT Gandhinagar to train students and researchers in professional scientific typesetting; organized week-long workshops attended by hundreds of students and faculty.